Shiv Nadar University, Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 03/08/2026 (Week 3) **Continuous Lab Evaluation-4** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Topics: 8051 Tool Chain setup – Development Board – AT89S52

Strictly No to using **Arduino APIs, HAL libraries, or RTOS services!**

Supporting Documents: **1.** Board Schematics 2. Pin diagram of 8051

Hardware Debug: Multi meter.

Human-Machine Interface using Matrix Keypad, LEDs, and Seven-Segment Display

Problem Statement:

Design a simple embedded user interface in which the keypad acts as an input device, the seven-segment display provides visual feedback, and the LEDs indicate different operating modes.

Whenever a valid key is pressed:

1. Display the key value on the seven-segment display.
2. Execute the corresponding LED function.

Functional Requirements

Key Press	Seven-Segment Display	LED Operation
1	1	LED1 ON
2	2	LED2 ON
3	3	LED3 ON
4	4	LED4 ON
5	5	Running LED Pattern
6	6	Alternate LEDs (1010 ↔ 0101)
7	7	Binary Counter (0–15)
8	8	All LEDs Blink
9	9	All LEDs OFF
0	0	Reset System

The program shall consist of separate functions:

```
void keypad_scan(void);
void display_digit(unsigned char value);
void led_control(unsigned char key);
void running_led(void);
void binary_counter(void);
void alternate_led(void);
void blink_all(void);
```

Deliverable form: Git repo